

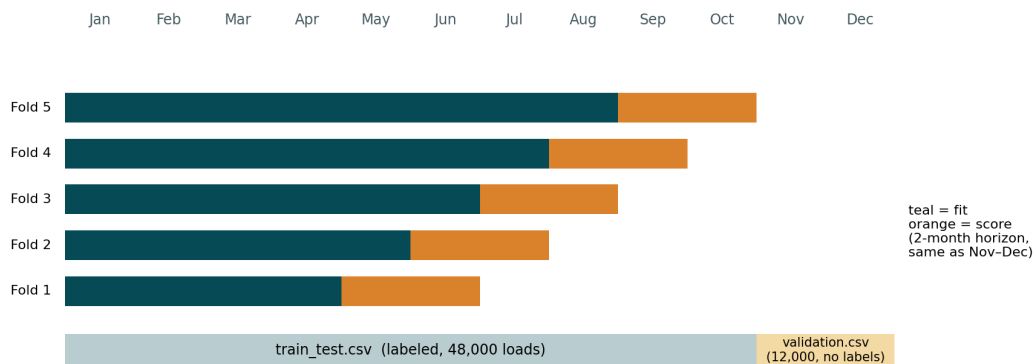
Freight Rate Prediction: Validation and Split Approach

Code, run instructions, dependencies and the full walkthrough notebook are in the accompanying GitHub repository.

1. The split decision

The validation set is a forecast, so validation must be a forecast. `train_test.csv` covers 2025-01-01 to 2025-10-31; `validation.csv` covers 2025-11-01 to 2025-12-31, with no date overlap. A random train/test split lets the model see days on both sides of each test row, which the real task never allows. I therefore used **rolling-origin (expanding-window) validation**: fit on everything before a cutoff, score the following two months, and repeat for five cutoffs (May to Sep). The two-month horizon matches the Oct-to-Nov gap of the real task. Every step (label cleaning, city effects, market context, model weights) is refit inside each fold.

I checked how much a random split would have misled me: the same plain LightGBM scores a MAE of about \$91 under shuffled 5-fold CV but \$120 under forward-in-time CV, so a shuffled split flatters the model by roughly a quarter. All figures below are means over the five forward folds. I compared model designs on these same folds, so they are slightly optimistic; the held-out Nov-Dec labels are the only true test.



2. Data exploration and quality issues

Corrupted labels (1.4%). After accounting for distance, equipment, weight, market, trend and city, typical error is about 1.3%. Yet 677 of 48,000 training rows (1.41%) sit far from any sensible value: 340 spikes (median 3.3x the expected rate) and 337 drops (median 0.28x). There is a clean gap between the populations (no kept row is off by more than 0.14 in log terms, no flagged row by less than 0.80), and the flagged share is about the same across equipment types, distance bands and months, so they are random corruption unrelated to any feature. They are removed from *training* only; evaluation labels are never filtered, and I report both all-row and "core" metrics (rows with plausible labels).

Other issues. 292 training weights are negative and look like sign flips (magnitudes are normal), so I use the absolute value. 300 weights are missing (LightGBM handles NaN natively). 374 `market_index` values are missing and are filled from that day's level. 48 loads sit at a 70-mile distance floor. Eight cities appear only in validation (Chicago, Charlotte, Knoxville, Norfolk, San Diego, Jackson, Allentown, Laredo), touching 12% of its rows. Coordinates are consistent with distance (correlation about 1.0) so they add no pricing information beyond it; I price with distance and use coordinates only to place cities regionally.

Structure found. (a) `log(distance)` and equipment dominate; rate per mile falls with distance. (b) `market_index` is effectively a *daily* series (98% of its variance is between days) with a weekly cycle and a few regime shifts; `quote_signal` is mostly per-load noise, so only its daily level is useful. (c) After removing load shape, the daily residual is explained at R-squared 0.92 by the log market level, the daily quote level and a **linear trend of about +0.7% per month**. (d) City effects are small (roughly -3.7% to +3.2%) and mostly smooth in geography (Northeast and Great Lakes low, Texas and Oklahoma high).

3. Model and why

$\log(\text{rate}) = \text{GBM}(\log \text{ distance, weight, equipment}) + \text{linear}(\log \text{ daily market level, row deviation from it, daily quote level, months elapsed, city effect of pickup and delivery})$, fit by backfitting on cleaned labels. Unseen cities get a k-nearest-neighbour estimate of their effect from coordinates. The reason for a hybrid rather than a single GBM is extrapolation: trees cannot continue a trend past the last training date, and the validation window lies entirely after it. The linear terms carry what must extrapolate; the GBM carries the non-linear load-shape effects.

Model (mean of 5 forward folds)	MAE (\$)	MAPE %	Median APE %	Core MAPE %	Median bias %
Median \$/mile baseline	221	10.46	6.87	7.45	-2.22
Plain LightGBM	120	5.03	2.28	2.81	-1.79
Hybrid (final)	105	4.51	1.92	2.22	+0.78

The hybrid beats the plain LightGBM on all five folds, cutting dollar error by 12% and core MAPE by 21%, and halves the error of a per-mile rule of thumb. The plain model under-forecasts progressively (median error drifting from about -1% to -3%), the signature of a tree that cannot continue the drift. Core metrics exclude the corrupted labels, which cause about half of the dollar error in the last fold.

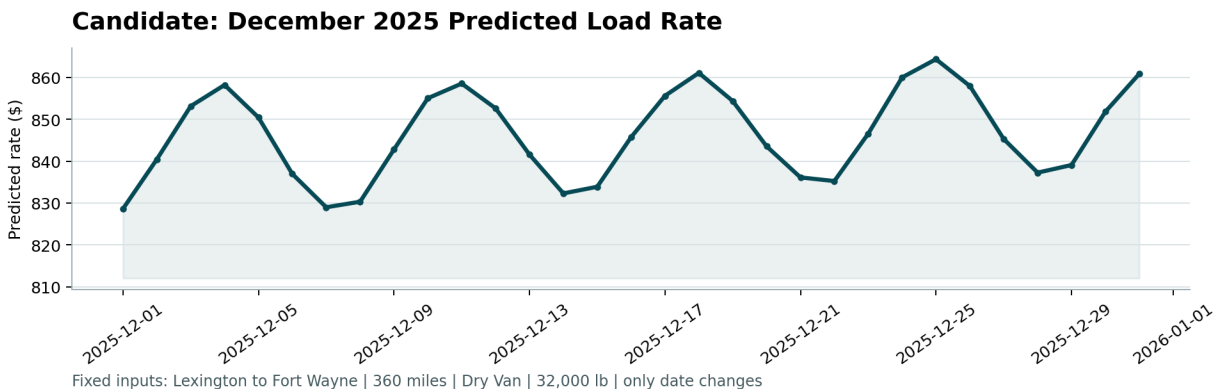
4. Unseen cities

To test the kNN fallback I hid 8 training cities, refit, and forecast Sep-Oct on loads touching them:

Scenario	MAPE %	Core MAPE %	Core log-MAE
Cities seen in training	4.02	2.09	0.0210
8 cities unseen (kNN fallback)	4.19	2.26	0.0230

Losing a city's history costs about 0.2 points of core MAPE, so the fallback is adequate for the 12% of validation rows affected.

5. Fixed December prediction chart (produced by the provided score.py)



Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb; only the date changes. The chart file has no market_index or quote_signal, so both are taken from that date's level in validation.csv (feature columns only, never labels). The weekly wave follows the market index and the drift is the fitted trend. Dec 25 falls on the weekly peak and is priced as an ordinary Thursday; the model has no holiday knowledge.

6. Limitations

(1) The trend is extrapolated two months past the data; the forward folds support it, but December seasonality cannot be learned from January to October, and the trend is partly entangled with market regime shifts in one year of data. (2) About 1.4% of the final validation labels are probably corrupted too; they will dominate any squared-error metric and no model can predict them. Training on cleaned labels targets the typical rate, which suits MAE or MAPE. (3) Lane-specific effects are not modelled. (4) Loads at the 70-mile distance floor are over-predicted by about 7% (small sample).